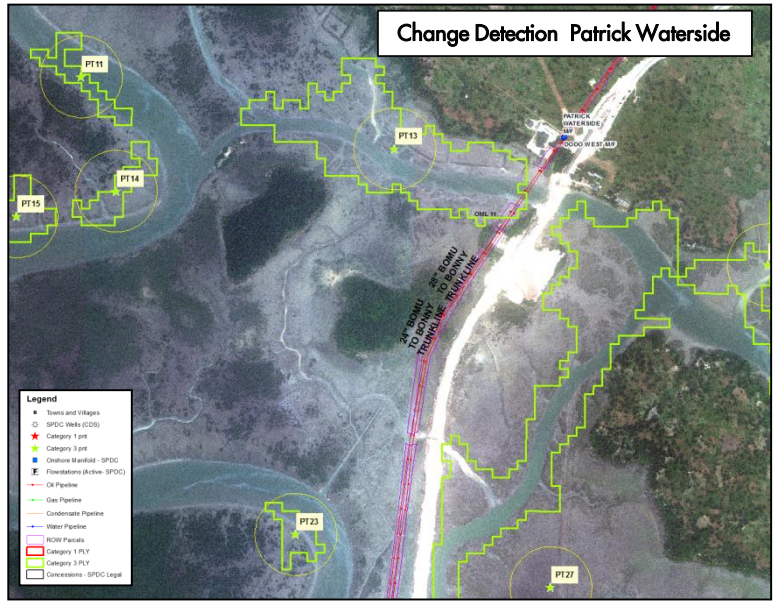
Title: Radar Satellite Imagery Environmental Change Detection Report		**Alert:** LOW
Date Issued: 3rd April,2019	**Geonatics Team: Remote Sensing Analyst**: Oyetunde Simisola, **Reviewed By:** Obowu Chituru, **Head Geoinformation Management**: Bisike-Ojiako Chuks	**Recipients:** Ojiako Uche UPO/G/UPR, Unutaro Tobore Grace UPO/G/UPR

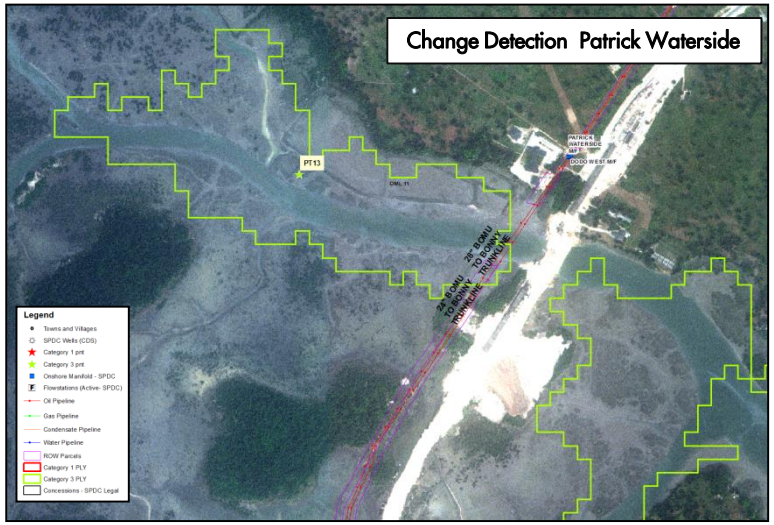
List of Change Detection Points and Attributes

S/N	Northings (Minna/Nigeria Mid Belt)	Eastings (Minna/Nigeria Mid Belt)	Longitude (WGS84)	Latitude (WGS84)	Alert Category	Alert Level	Change Detection Date	Remarks
PT 11	67651.96	531642.37	7.24728	4.61152	Category 3	Low	20190305-20190329	Changes of water coverage (e.g. tidal, river, flood) as well as wetland related changes
PT 13	67388.42	532791.86	7.257644	4.609155	Category 3	Low	20190305-20190329	Changes of water coverage (e.g. tidal, river, flood) as well as wetland related changes
PT 14	67232.77	531770.69	7.248444	4.607731	Category 3	Low	20190305-20190329	Changes of water coverage (e.g. tidal, river, flood) as well as wetland related changes
PT 15	67139.51	531406.53	7.245163	4.606882	Category 3	Low	20190305-20190329	Changes of water coverage (e.g. tidal, river, flood) as well as wetland related changes
PT 17	66963.37	534161.19	7.269991	4.605333	Category 3	Low	20190305-20190329	Changes of water coverage (e.g. tidal, river, flood) as well as wetland related changes
PT 21	66399.65	534480.79	7.272881	4.60024	Category 3	Low	20190305-20190329	Changes of water coverage (e.g. tidal, river, flood) as well as wetland related changes
PT 23	65974.88	532429.09	7.254397	4.596366	Category 3	Low	20190305-20190329	Changes of water coverage (e.g. tidal, river, flood) as well as wetland related changes
PT 27	65780.41	533364.42	7.262829	4.594622	Category 3	Low	20190305-20190329	Changes of water coverage (e.g. tidal, river, flood) as well as wetland related changes

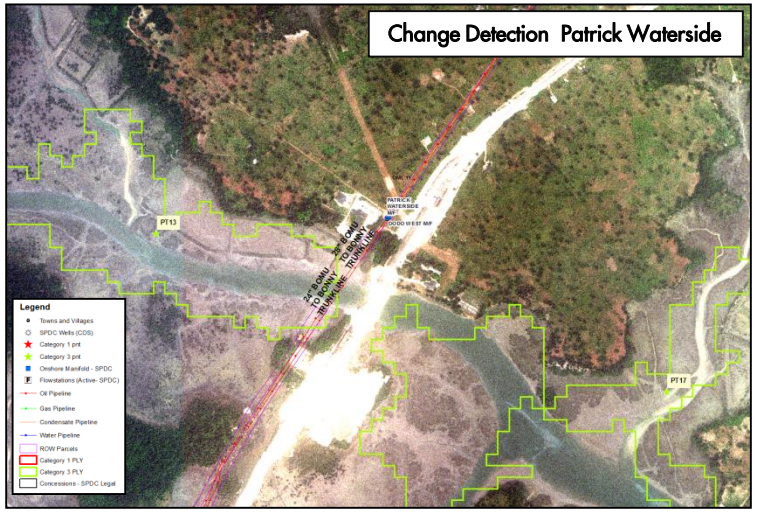
Note: *Points are within 150m sensitivity zone of the centroid of the identified change areas.*



Clip 1: Full view of change detection



Clip 2: View of pt13 change detection



Clip 3: View of pt13 &17 change detection

Summary

Concession : OML 11

Local Government: Bonny

Settlements: NTR

Wells: Bodo West-004, Bodo West-003

Flow Stations: Nil

Manifolds: Patrick Waterside M/F, Dodo North M/F, Bodo West M/F

Pipelines: 24" Bomu to Bonny Trunk line, 28" Bomu to Bonny Trunk line

Alert Category Description

- **Category 1 Targets** - metallic man-made features such as vehicle movement, ship traffic or structures.
- **Category 2 Canopy/Surface** - tree canopy changes/ disturbance) and surface disturbance.
- **Category 3 Low Backscatter** - e.g. tidal, river, flood) as well as wetland related changes.

Remote sensing Analysis Result

- Possible changes are within 150m radius of detected alert points as analyzed by radar satellite imagery acquired on 05/03/2019 and 29/03/2019.
- Category 3 changes identified within this area. Changes possibly related to tidal, river, flooding as well as wetland related changes.
- All points are category 3 change which is most likely changes in water coverage (e.g. tidal, river, flood) as well as wetland related changes.
- All alert level are low.
- No action required.

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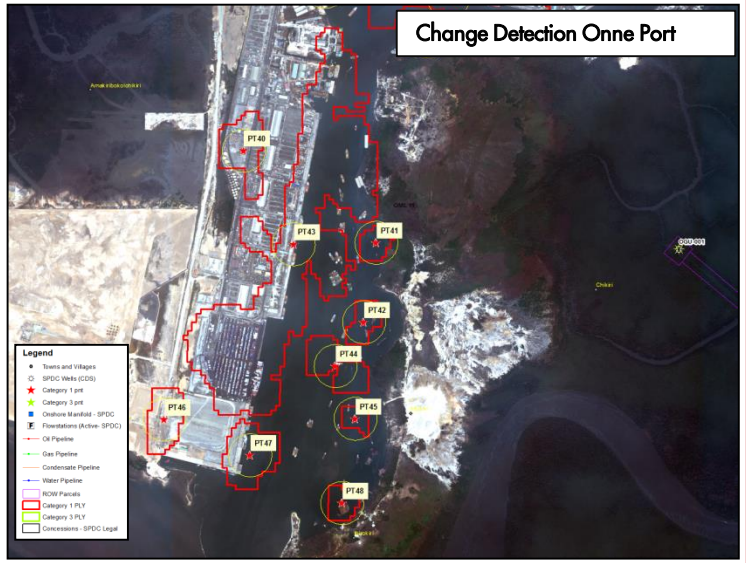
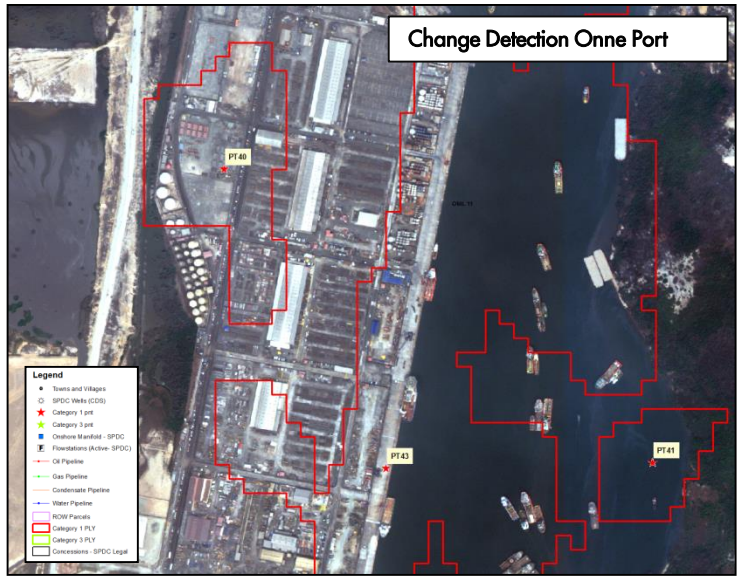
List of Change Detection Points and Attributes

S/N	Northings (Minna/Nigeria Mid Belt)	Eastings (Minna/Nigeria Mid Belt)	Longitude (WGS84)	Latitude (WGS84)	Alert Category	Alert Level	Change Detection Date	Remarks
PT 40	75850.79	533755.61	7.152743	4.685487	Category 1	Low	20190305-20190329	Vehicle movement, ship ,changes to or the addition of new structures.
PT 41	75208.79	522095.13	7.16111	4.679698	Category 1	Low	20190305-20190329	Vehicle movement, ship ,changes to or the addition of new structures.
PT 42	74653.68	522008.94	7.160343	4.674676	Category 1	Low	20190305-20190329	Vehicle movement, ship ,changes to or the addition of new structures.
PT 43	75197.75	521517.85	7.155908	4.679588	Category 1	Low	20190305-20190329	Vehicle movement, ship ,changes to or the addition of new structures.
PT 44	74346.83	521809.70	7.158553	4.671898	Category 1	Low	20190305-20190329	Vehicle movement, ship ,changes to or the addition of new structures.
PT 45	73978.88	521949.04	7.159815	4.668573	Category 1	Low	20190305-20190329	Vehicle movement, ship ,changes to or the addition of new structures.
PT 46	73972.59	520613.31	7.147777	4.668493	Category 1	Low	20190305-20190329	Vehicle movement, ship ,changes to or the addition of new structures.
PT 47	73723.70	521213.69	7.153192	4.666253	Category 1	Low	20190305-20190329	Vehicle movement, ship ,changes to or the addition of new structures.
PT 48	73389.66	521856.09	7.158988	4.663243	Category 1	Low	20190305-20190329	Vehicle movement, ship ,changes to or the addition of new structures.

Note: Points are within 150m sensitivity zone of the centroid of the identified change areas.



Clip 1: Full view of change detection



Clip 2: View of Pt40,Pt41,Pt43 change detection

Summary

Concession :OML11

Local Government: Ogu/Bolo

Settlements: Ibirokiri, Chikiri, Uwogolo, Amakiribokoiohikiri, Amakiriboko

Wells: Ogu-001

Flow Stations: Nil

Manifolds: Nil

Pipelines: Nil

Alert Category Description

- **Category 1 Targets** - metallic man-made features such as vehicle movement, ship traffic or structures.
- **Category 2 Canopy/Surface** - tree canopy changes/ disturbance) and surface disturbance.
- **Category 3 Low Backscatter** - e.g. tidal, river, flood) as well as wetland related changes.

Remote sensing Analysis Result

- Possible changes are within 150m radius of detected alert points as analyzed by radar satellite imagery acquired on 05/03/2019 and 29/03/2019.
- Category 1 changes identified within this area. Changes possibly related to vehicle movement, ship, changes to or the addition of new structures .This area is not within our primary area of interest. But alert is for information only.
- All alert level are low.
- No action required.

Clip 2: View of Pt42,Pt44,Pt45 change detection